

- 运动1a

1. 参考系

$$|dr| \neq dr$$

$$\frac{d(A)}{dt} = 2A \frac{dA}{dt}$$

$$\frac{d\hat{A}}{dt} = \frac{d\hat{A}}{dt} \cdot A + \frac{dA}{dt} \cdot \hat{A}$$

$$\frac{dA}{dt} = \lim_{\Delta t \rightarrow 0} \frac{2s \sin(100/2)}{\Delta t} = \frac{d\phi}{dt} \quad (\phi \rightarrow 0)$$

$\hat{a}$ 为常量 - 垂直

$\hat{a}$ 向定 - 极

轨迹确定 - 自然

△ 自然 - 单位矢量方向变化

$$\dot{\mathbf{x}} \cdot \dot{\mathbf{x}} = \dot{\mathbf{x}} \cdot \left(\frac{d\hat{\mathbf{x}}}{dt} \cdot \hat{\mathbf{x}} + \frac{dx}{dt} \cdot \hat{\mathbf{x}} \right)$$

$$\text{若 } \frac{dx}{dt} = 0$$

$$\text{可证 } \dot{\mathbf{x}} \cdot \dot{\mathbf{x}} = 0$$

△ 极

$$v_r(\text{径向}) = \frac{dr}{dt} = \dot{r}$$

$$\dot{\mathbf{r}} \cdot \hat{\mathbf{r}}$$

$$v_\theta(\text{横向}) = r \cdot \frac{d\theta}{dt} = r \cdot \dot{\theta}$$

$$r \cdot \dot{\theta} \cdot \hat{\theta}$$

$$\dot{\theta} = \omega \quad \dot{r} = v$$

开普勒定律

相同 - 面积速度 $\frac{d^2}{dt^2}$

二. 运动1b

沙尔定理. 位 - 位移 = 平动 + 转动

基点不同 速度 { 平动 不同
转动 相同

参考系间平动 $r \cdot v \cdot a = r \cdot v \cdot a \text{ S系} + \text{相对}$

转动 $v_{\text{相对}} = v_{\text{相对}} + \omega \times r$

$$\rho(\text{曲率半径}) = \frac{v_\perp^2}{a_n}$$

$$\vec{a}_\theta(t) = (2\dot{r} \cdot \dot{\theta} + r \cdot \ddot{\theta}) \cdot \hat{\theta} = \frac{1}{r} \frac{d}{dt} (r^2 \dot{\theta})$$

$$\vec{a}_r(t) = (\ddot{r} - r \dot{\theta}^2) \cdot \hat{r}$$

$$\vec{v} = \vec{\omega} \times \vec{R}$$

$$\begin{aligned} \vec{a} &= \frac{d\vec{\omega}}{dt} \times \vec{R} + \frac{d\vec{R}}{dt} \times \vec{\omega} \\ &= \frac{d\vec{\omega}}{dt} \times \vec{R} + \vec{\omega} \times (\vec{\omega} \times \vec{R}) \end{aligned}$$

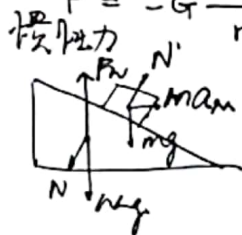
$$\begin{aligned} \vec{v} &= \vec{v}' + \vec{\omega} \times \vec{r} \\ \vec{a} &= \vec{a}' + \vec{\omega} \times \vec{v}' \\ + \vec{a} &= \frac{d\vec{\omega}}{dt} \times \vec{r} + \vec{\omega} \times (\vec{\omega} \times \vec{r}) \\ &\parallel \\ \vec{a} &= \vec{a}' + 2\vec{\omega} \times \vec{v}' + \left(\frac{d\vec{\omega}}{dt} \times \vec{r} + \vec{\omega} \times (\vec{\omega} \times \vec{r}) \right) \\ &\text{相对} \quad \text{科} \quad \text{牵连} \end{aligned}$$



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三. 运动与力

$$\vec{F} = -G \frac{m_1 m_2}{r^2} \vec{e}_r$$



$$mg \sin \theta + m a_{\text{cm}} \cos \theta = m a$$

$$mg \cos \theta = N + m a_{\text{cm}} \sin \theta$$

$$N \cos \theta + m g = F_f$$

$$N \sin \theta = m a_{\text{cm}}$$

惯性力在转换后莫忘加

转动中 惯性力

$$\vec{F} = m \vec{\omega}^2 \vec{r} - 2 m \vec{\omega} \times \vec{v}$$

火箭速度公式

$$m dv = -u dm$$

$$v_f = v_i + u \ln \frac{m_i}{m_f}$$

四. 动量, 角动量

$$I = Ft$$

$$F = ma$$

$$I = \Delta m v$$

$$F = -G \frac{m_1 m_2}{r^2} \vec{e}_r$$

$$\text{表面张力 } F = \gamma l$$

$$m \Delta r l = m g \cdot "2" - \text{膜}$$

$$D: \text{ 泡 } P_0 \pi R^2 + 2 \gamma \cdot 2 \pi R = P \cdot \pi R^2$$

$$P_{\text{in}} - P_0 = \frac{2 \gamma \cdot 2 \pi R}{\pi R^2} = \frac{4 \gamma}{R}$$

终端速度 ($f = kv$)

$$v_T = \frac{mg - f_{\text{其他}}}{k}$$

$$\frac{dv}{dt} = \frac{k(v_T - v)}{m}$$

积分

$$v = v_T (1 - e^{-\frac{k}{m} t})$$

$$F_{\text{向}x} = G \sigma m_{\text{月}} \left(\frac{r - R \cos \theta}{(R^2 + r^2 - 2Rr \cos \theta)^{3/2}} - \frac{1}{r^2} \right)$$

$$\approx \frac{2 G \sigma m_{\text{月}}}{r_{\text{地}}^3} \cdot R_{\text{地}} \cos \theta$$

$$F_{\text{向}y} = - \frac{G \sigma m_{\text{月}}}{r_{\text{地}}^2} R_{\text{地}} \sin \theta$$

$$\text{洛希极限: } r = \left(\frac{2 P_{\text{月}}^{1/3} R_{\text{地}} m_{\text{地}}}{\rho_{\text{月}}} \right)$$

$$g \approx a_g - a_{\text{地}} \times \cos \theta$$

傅科摆, 科里奥利力

$$\vec{\Omega} = -\vec{\omega}_r$$

$$\frac{T}{T_0} = \frac{\omega}{\Omega} = \frac{1}{\sin \theta}$$

哈密顿-薛定谔方程

$$m \frac{d\vec{v}}{dt} = (\vec{v} - \vec{v}_0) \frac{dm}{dt} + \vec{F}$$

质心

$$m_c \vec{v}_c = \sum m_i \vec{v}_i$$

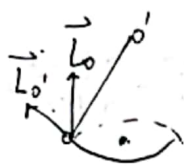
系点系动量守恒 $\rightarrow$ 质心匀速运动



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$\times \vec{p}$

质点同一运动 固定点(基点)选择不同
角动量可能改变



$$||\vec{L}|| = r p \sin \alpha \quad \text{kg m}^2/\text{s}$$

角动量定理

$$\vec{M} = \vec{r} \times \vec{F} = \frac{d\vec{L}}{dt}$$

$$\vec{r}_0 = \vec{r} \sin \theta \quad \text{力臂}$$

对轴的角动量定理

$$M_z = \frac{dL_z}{dt}$$

质点系角动量定理

$$\vec{M}_O = \frac{d\vec{L}_O}{dt}$$

$$\vec{L} = \vec{L} - \vec{L}_c$$

(质心对质心) (质心对质心) (质心对质心)

质心系

$$\sum m_i \vec{r}_i' = 0$$

$$\sum m_i \vec{v}_i' = 0$$

五. 功和能

△ 功: 力和力所作用的质点(或质元)的位移的标积

依赖参考系, 有正负

△ 动能定理: $\vec{F} \cdot d\vec{r} = d(\frac{1}{2}mv^2)$

$$= \frac{1}{2}m \cdot 2\vec{v} \cdot d\vec{v}$$

$$= m\vec{v} \cdot d\vec{v}$$

内力成对 但做功未必一致

△ 一对力的功 大小相等, 方向相反

$$dW_{12} = \vec{F}_1 \cdot d\vec{r}_1 + \vec{F}_2 \cdot d\vec{r}_2$$

$$= -\vec{F}_1 \cdot d\vec{r}_2 + \vec{F}_2 \cdot d\vec{r}_1$$

$$= \vec{F}_1 \cdot (d\vec{r}_1 - d\vec{r}_2)$$

2对1的元位移, 与参考系选取无关

△ 保守力: $\oint \vec{F} \cdot d\vec{r} = 0$

是否为保守力与参考系无关

△ 势能: 势能的减小等于保守内力的功

属于系统

不依赖参考系 依赖点

质点系动能定理

$$W_{\text{外}} + W_{\text{内}} = E_{k2} - E_{k1}$$

$$dW_{\text{对}} = \vec{F}_2 \cdot d\vec{r}_2$$

$$\text{万有引力势能 } E_p(r) = -\frac{G M m}{r} + C$$

mg 特例



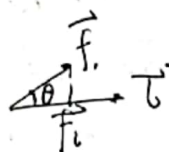
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△ 由势能求保守力

$$f_{保} = - \frac{\partial E_p}{\partial l}$$

$$F_{保} = - \text{grad} \cdot E_p$$

势能曲线 斜率为负 正 力为引力



注意：分析结果合理性。

① 量纲。

② 特殊情况。

△ 功能原理。

$$W_{外} + W_{内非} = \Delta E_k$$

$$W_{外} + W_{内非} = \Delta E_{机}$$

保守内力做功是 E_k 与 E_p 转化的方式和度量。

△ 有心运动

有心保守力场

E 守恒

L 守恒

B 守恒

$$\vec{B} = \vec{v} \times \vec{L} = G M m \frac{\vec{r}}{r}$$

$$\vec{r} \cdot \vec{B} = \frac{L^2}{m} - G M m r$$

$\vec{B}$ 沿通过焦点的对称轴，指向最近拱点。

圆轨道 $B=0$ $\parallel \vec{B} \parallel$ 正比于 $\frac{c}{a} = e$

弹性势能 $E_p(x) = \frac{1}{2} k x^2 + C$

$E_p(0) = 0$ $E_p(x) = \frac{1}{2} k x^2$

直角坐标系

$$\text{grad} \varphi = \nabla \varphi = \frac{\partial \varphi}{\partial x} \vec{i} + \frac{\partial \varphi}{\partial y} \vec{j} + \frac{\partial \varphi}{\partial z} \vec{k}$$

柱坐标系

$$\begin{cases} x = r \cos \varphi \\ y = r \sin \varphi \\ z = z \end{cases}$$

侧面图。



$$df = \frac{\partial f}{\partial r} \vec{e}_r + \frac{\partial f}{\partial \varphi} \frac{1}{r} \vec{e}_\varphi + \frac{\partial f}{\partial z} \vec{e}_z$$

球坐标系

$$df = \frac{\partial f}{\partial r} \vec{e}_r + \frac{\partial f}{\partial \varphi} \frac{1}{r} \vec{e}_\varphi + \frac{\partial f}{\partial \theta} \frac{1}{r \sin \theta} \vec{e}_\theta$$



$$(\vec{a} \times \vec{b}) \cdot \vec{c} = \vec{b} \cdot (\vec{a} \times \vec{c}) = \vec{c} \cdot (\vec{a} \times \vec{b})$$

$$\vec{B} = \vec{v} \times \vec{L} = G M m \frac{\vec{r}}{r}$$

$$(\vec{a} \times \vec{b}) \cdot \vec{c} = (\vec{c} \times \vec{a}) \cdot \vec{b} = \vec{a} \cdot (\vec{b} \times \vec{c})$$

$$r = \frac{p}{1 + e \cos \theta}$$

$$p = \frac{L^2}{G M m^2}$$

$$e = \frac{B}{G M m}$$

$$B = G M m \sqrt{1 + \frac{2 L^2 E}{G^2 M^2 m^3}}$$

$$e = \frac{B}{G M m} = \sqrt{1 + \frac{2 L^2 E}{G^2 M^2 m^3}}$$

$$p = \frac{L^2}{G M m^2}$$



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压缩.

$$m_1 v - m_1 u_1 = -I$$

$$m_2 v - m_2 u_2 = I$$

$$u_1 - u_2 = I \left(\frac{1}{m_1} + \frac{1}{m_2} \right)$$

$$u_1 - u_2 = I \cdot \frac{1}{\mu}$$

$$\left(\mu = \frac{m_1 m_2}{m_1 + m_2} \right)$$

$$\begin{cases} m_1 v_1 - m_1 v = -I \\ m_2 v_2 - m_2 v = I \end{cases}$$

$$v_2 - v_1 = I \left(\frac{1}{m_1} + \frac{1}{m_2} \right)$$

$$v_2 - v_1 = I \cdot \frac{1}{\mu}$$

$$\frac{v_2 - v_1}{u_1 - u_2} = \frac{I}{I} = e$$

$$\Delta E_k = \left(\frac{1}{2} \mu (u_1 - u_2)^2 - \frac{1}{2} \mu (v_1 - v_2)^2 \right)$$

$$= \frac{1}{2} \mu (1 - e^2) (u_1 - u_2)^2$$

若 m_2 静止

$$= \frac{1}{2} \mu (e^2 - 1) (u_1 - u_2)^2$$

$$\frac{\Delta E_{k2}}{E_{k1}} = \frac{\frac{1}{2} m_2 u_2^2}{\frac{1}{2} m_1 u_1^2}$$

$$= \frac{4 m_1 m_2}{(m_1 + m_2)^2}$$

$$= \frac{4 \frac{m_1}{m_2}}{(1 + \frac{m_1}{m_2})^2}$$

通过小球触地回弹系数 e

$$M \gg m$$

$$v = -eu$$

$$e = \frac{|v|}{|u|} = \sqrt{\frac{h}{h}}$$

△两体问题

变两体为单体

$$e = \frac{J}{I} \in (0, 1)$$

$$v_1 = \frac{m_1 - e m_2}{m_1 + m_2} u_1 + \frac{(1+e)m_2}{m_1 + m_2} u_2$$

$$v_2 = \frac{m_2 - e m_1}{m_1 + m_2} u_2 + \frac{(1+e)m_1}{m_1 + m_2} u_1$$

$$\Delta E_k = \frac{1}{2} \mu (e^2 - 1) (u_1 - u_2)^2$$

$$\frac{m_1}{m_2} = 1 \text{ 时 } \frac{\Delta E_k}{E_{k1}} \text{ 最大}$$

故可指导对撞

两质点 $e=0$ 完全非弹性

碰后速度为质心速度

通过

质心运动定理

$$M \cdot \sum m_i \cdot a_{ci} = 0$$

$$\mu \cdot \ddot{r}_{12} = f_{12} \quad \text{或} \quad m \cdot \ddot{r}_{12} = f_{12}$$

$$\mu \cdot \ddot{r}_{21} = f_{21} \quad \dots$$



扫描全能王 创建

太阳参考系 $m \cdot \frac{d^2 r}{dt^2} = f$

$M \cdot \frac{d^2 R}{dt^2} = f'$

$\mu f = \frac{d^2(r-R)}{dt^2} \quad (r-R = r')$
太阳系

$f = \mu \cdot \frac{d^2 r'}{dt^2}$ 行星运动方程.
↓
m a.

两体化为一休 r' .

$M \gg m \quad \mu \approx m$

$f = m \frac{d^2 r'}{dt^2}$

相对偏差 $\Delta = \frac{m-\mu}{\mu} = \frac{m}{M} \approx 0$

$\frac{d^2 r}{dt^2} = \frac{d^2 R}{dt^2} + \frac{d^2 r'}{dt^2}$

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大物B(1)期末·一·振动与波、

简谐振动·无阻尼自由简谐振动 (SHM)

△判据: ① 受力 $F = -kx$

(无阻尼自由) ② 微分方程 $\frac{d^2x}{dt^2} + \omega^2 x = 0$

③ 能量特征: $\begin{cases} E_{\text{总}} = \text{const.} \quad (1) \\ E_p = \frac{1}{2} k x^2 \quad (2) \end{cases}$
(平衡位置 $x_p = 0$)
(2) or $E_p = E_k = \frac{1}{2} k A^2 \propto A^2$

△特征量: →

△表示法: ① 解析式:

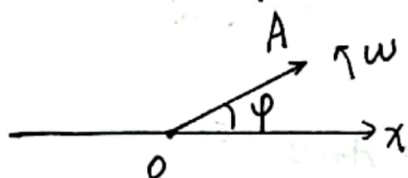
$$x = A \cos(\omega t + \varphi)$$

(求一次导. 加 $\frac{\pi}{2}$) $v = \frac{dx}{dt} = \omega A \cos(\omega t + \varphi + \frac{\pi}{2})$

$$a = \frac{dv}{dt} = -\omega^2 A \sin(\omega t + \varphi + \frac{\pi}{2}) = \omega^2 A \cos(\omega t + \varphi + \pi)$$

② 振动曲线

△矢量对应(相位图):



△复数对应:

$$x = A \cos(\omega t + \varphi)$$

$$\tilde{x} = A e^{i(\omega t + \varphi)}$$

相位差 $\Delta\varphi = \varphi_2 - \varphi_1$

$\begin{cases} \Delta\varphi = 2k\pi & \text{同相} \\ \Delta\varphi = (2k+1)\pi & \text{反相} \\ \text{其它值 } |\Delta\varphi| \leq \pi & \Delta\varphi > 0 \text{ 超前} \\ & \Delta\varphi < 0 \text{ 落后} \end{cases}$

角频率 $\sqrt{\frac{k}{m}} (Hz) = \omega$

频率 $\nu = \frac{\omega}{2\pi} (Hz)$

周期 $T = \frac{1}{\nu} = \frac{2\pi}{\omega}$

振幅 $A = \sqrt{x_0^2 + \frac{v_0^2}{\omega^2}} = \sqrt{\frac{2E}{k}}$
($E = \frac{1}{2} k A^2$)

相位. $\tan \varphi = -\frac{v_0}{\omega x_0}$
量纲为1.



复摆

净力矩

$$-mgl \sin \theta \approx -mgl\theta$$

(0.1.时) $-mgl\theta = m \cdot \frac{d^2 \theta}{dt^2}$

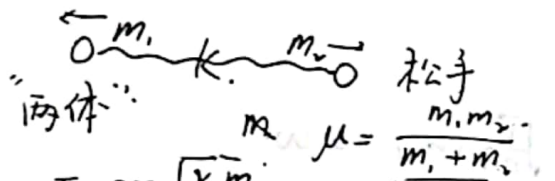
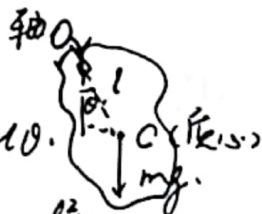
$$= J \cdot \frac{d^2 \theta}{dt^2}$$

$$\frac{d^2 \theta}{dt^2} + \frac{mgl}{J} = 0$$

$$\frac{d^2 x}{dt^2} + \omega^2 x = 0$$

$$\Rightarrow \omega = \sqrt{\frac{mgl}{J}}$$

单摆 $J = ml^2$



两体 $\mu = \frac{m_1 m_2}{m_1 + m_2}$
 $T = 2\pi \sqrt{\frac{\mu}{k}} = 2\pi \sqrt{\frac{m_1 m_2}{k(m_1 + m_2)}}$

阻尼振动

$$f = -r v$$

阻力系数

$$m \frac{d^2 x}{dt^2} = -r \frac{dx}{dt} - kx$$

(二阶线性常系数)

$$\ddot{x} + 2\beta \dot{x} + \omega_0^2 x = 0$$

新增项

$$\beta = \frac{r}{2m}$$

$$\omega_0 = \sqrt{\frac{k}{m}}$$

过阻尼 $\beta > \omega_0$

临界阻尼 $\beta = \omega_0$

欠阻尼 $\beta < \omega_0$

过程中
T↑ 但不变化

$$(\omega = \sqrt{\omega_0^2 - \beta^2})$$

复摆角频率 (0.1.时)

$$\omega = \sqrt{\frac{mgl}{J}}$$

单摆 $\omega = \sqrt{\frac{g}{l}}$

$$\ddot{x} + 2\beta \dot{x} + \omega_0^2 x = 0$$

$$\beta = \frac{r}{2m}$$

$$\omega_0 = \sqrt{\frac{k}{m}}$$

振幅 $A = A_0 \cdot e^{-\beta t}$

时间常数 $\tau = \frac{1}{\beta} = \frac{1}{\frac{r}{2m}} = \frac{2m}{r}$

Q $Q = \frac{2\pi T}{T} = \omega \tau = \frac{\omega \times 2m}{r}$

$\omega = \sqrt{\omega_0^2 - \beta^2}$

通解

过阻尼 $\beta > \omega_0$ $x(t) = A_1 e^{-(\beta - \omega_0)t} + A_2 e^{-(\beta + \omega_0)t}$
 $\beta_0 = \sqrt{\beta^2 - \omega_0^2}$

临界阻尼 $\beta = \omega_0$ $x(t) = (A_1 + A_2 t) e^{-\beta t}$

欠阻尼 $x(t) = A_0 \cdot e^{-\beta t} \cos(\omega_0 t + \varphi_0)$
 $\omega = \sqrt{\omega_0^2 - \beta^2}$



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$$A_{\text{阻}} = \int_0^T f v dt = \int_0^T -r v^2 dt = -4\beta \int_0^T \frac{1}{2} m v^2 dt$$

$$\bar{E}_k = \frac{1}{T} \int_0^T \frac{1}{2} m v^2 dt$$

$$\beta \ll \omega_0, \quad \bar{E}_k \approx \bar{E}_p$$

$$\bar{E} = \bar{E}_k + \bar{E}_p \longrightarrow$$

受迫振动.

$$F = H \cos \omega t.$$

$$h = \frac{H}{m}.$$

稳态时 $\omega = \omega_{\text{驱}}$.

$$\beta \ll \omega_0, \quad \text{当 } \omega = \omega_0 \quad \text{时, } A \text{ 达 } \text{max}$$

$$\ddot{x} + 2\beta \dot{x} + \omega_0^2 x = h \cos \omega t$$

特解 $x = A \cos(\omega t + \varphi)$

$$\Rightarrow x = A e^{i(\omega t + \varphi)}$$

但共振.

速度的振幅与相位.

$$A_{\text{阻}} = -4\beta T \bar{E}_k.$$

$$\beta \ll \omega_0.$$

$$A_{\text{阻}} = -2\beta T \bar{E}$$

$$|\Delta E| = 2\beta T \bar{E}$$

$$\frac{|\Delta E|}{\bar{E}} = 2\beta T \quad \text{系统性质.} \quad \beta \ll \omega_0$$

$$Q = 2\pi \frac{\bar{E}}{|\Delta E|} = \frac{\pi}{\beta T} = \frac{\omega}{2\beta} \approx \frac{\omega_0}{2\beta}$$

无量纲

$$= 2\pi \frac{T}{\tau}$$

鸣响时长 $\sim T$. (时间常数)

受迫振动 $F = H \cos \omega t$

$$h = \frac{H}{m}$$

$$\frac{E}{E_0} = e^{-2\beta t}.$$

振幅相位公式

$$A = \frac{h}{[(\omega_0 - \omega)^2 + 4\beta^2 \omega^2]^{1/2}}$$

$$\varphi = \arctan \frac{-2\beta \omega}{\omega_0^2 - \omega^2}$$

$$\omega_r = \sqrt{\omega_0^2 - 2\beta^2}$$

$$A_r = \frac{h}{2\beta \sqrt{\omega_0^2 - \beta^2}}$$

$$\beta \ll \omega_0, \quad \omega_r \approx \omega_0$$

$$A_r = \frac{h}{2\beta \omega_0}$$

$$\varphi_r = \frac{\pi}{2} + \varphi$$

$$\left\{ \begin{aligned} v &= \omega A = \frac{\omega h}{[(\omega_0 - \omega)^2 + 4\beta^2 \omega^2]^{1/2}} \\ \varphi_v &= \frac{\pi}{2} - \arctan \frac{2\beta \omega}{\omega_0^2 - \omega^2} \end{aligned} \right.$$



驱动力. 平均功率

$$P = \frac{1}{T} \int_0^T f_{阻} \cdot v dt = \frac{\omega}{2\pi} \int_0^{2\pi} H V \cos \omega t \cos \omega t + \varphi dt.$$

$$= \frac{1}{2\pi} \int_0^{2\pi} H V \cos \omega t [\cos(\omega t + \varphi) + \cos(\omega t - \varphi)] dt.$$

$$= \frac{1}{2} H V \cos \varphi.$$

速度共振: 一定条件下, 速度之“振幅” ωA 最大
同向振动合成 { 矢量图解法
同向合成 (振动方向) 复数法,
• 同频率振动相位差不变性

$$\omega_1 \approx \omega_2, \gg |\omega_1 - \omega_2| \text{ 拍}$$

$$\nu = |\omega_1 - \omega_2| \text{ 拍频}$$

$$x_1 = A \cos(\omega_1 t + \varphi)$$

$$x_2 = A \cos(\omega_2 t + \varphi)$$

$$x = 2A \cos \frac{\omega_2 - \omega_1}{2} t \cos \left(\frac{\omega_1 + \omega_2}{2} t + \varphi \right)$$

$$\omega_2 \approx \omega_1, \quad \omega_2 - \omega_1 \ll \omega_2 + \omega_1,$$

$$x = A(t) \cos \bar{\omega} t$$

$$A(t) = 2A \cos \left(\frac{\omega_2 - \omega_1}{2} t \right) \text{ 小, 随 } t \text{ 缓慢变化}$$

$$\cos \bar{\omega} t = \cos \left(\frac{\omega_1 + \omega_2}{2} t \right) \rightarrow \text{随 } t \text{ 快速变化}$$

$$\nu = \frac{2\pi}{T} = \frac{\omega_2 - \omega_1}{2}$$

合成

$$x = A_1 \cos(\omega t + \varphi_1)$$

$$y = A_2 \cos(\omega t + \varphi_2)$$

$$\frac{x^2}{A_1^2} + \frac{y^2}{A_2^2} - 2 \frac{x}{A_1} \frac{y}{A_2} \cos(\varphi_2 - \varphi_1) = \sin^2 \varphi_2 \varphi_1$$

椭圆. 形状决定于 $A_1, A_2, \Delta \varphi$

$$\frac{\omega_1}{\omega_2} = \frac{m}{n} \quad (m, n \in \mathbb{N}^+)$$

合成轨迹, 稳定的闭合曲线, 李萨如图.

谐波分析 付里叶分解 方波分解为 $\omega_0, 3\omega_0, 5\omega_0$

$$T = \frac{1}{2\beta}$$

$$Q = 2\pi \frac{T}{\tau}$$

$$\omega = \sqrt{\omega_0^2 - \beta^2}$$

$$\bar{P} = \frac{1}{2} H V \cos \varphi_r$$

$$\omega_r = \omega_0$$

$$\gamma_r = \frac{1}{2}\beta$$

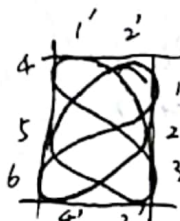
$$\varphi_{r,r} = 0$$

速度与驱动力同相 正功 E 输入 α $\cos \alpha$

$$x = 2A \cos \frac{\omega_2 - \omega_1}{2} t \cdot \cos \left(\frac{\omega_1 + \omega_2}{2} t + \varphi \right)$$

$$A = \sqrt{A_1^2 + A_2^2 + 2A_1 A_2 \cos(\varphi_2 - \varphi_1)}$$

$$\varphi = \arctan \frac{A_1 \sin \varphi_1 + A_2 \sin \varphi_2}{A_1 \cos \varphi_1 + A_2 \cos \varphi_2}$$



$$\text{李萨如图. } \frac{\omega_x}{\omega_y} = \frac{\nu_x}{\nu_y} = \frac{x \text{ 达到最大次数}}{y \text{ 达到最大次数}}$$

$$\text{谐波分析 } x(t) = \frac{a_0}{2} + \sum_{k=1}^{\infty} [A_k \cos(k\omega t + \varphi_k)]$$

$k=1$ 基频 ω $\rightarrow$ 音调
 $k=2$ 二次谐波
 $k=3$ 三次谐波
高次谐波, 决定音色.

$$\beta \ll \omega_0 \text{ (弱阻尼)} \Delta \omega = 2\beta$$

$$Q = \omega T = \frac{\omega_0}{2\beta} = \frac{\omega_0}{\Delta \omega} = \frac{\text{固有频率}}{\text{谱线宽度}}$$



扫描全能王 创建

大学物理学B(1) 波

行波, 简谐波

形成机械波条件

- 波源
- 弹性介质

特征量

波速 u

周期 T

频率 $\nu = \frac{1}{T}$

角频率 $\omega = 2\pi\nu$

波长 $\lambda = uT$

行波 $y = f(x \pm \frac{x}{u})$

$$y(x+\Delta x, t+\Delta t) = y(x, t)$$

波函数 $x=0$ $y(0, t) = A \cos \omega t$

$u = \frac{dx}{dt}$ 相速 (简谐波波速即相速)

沿波方向每过 λ 距离, 相位落后 2π

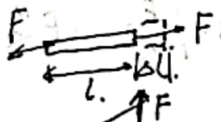
$\begin{cases} x \text{ 一定, } y \sim t, & x \text{ 处振动图像} \\ t \text{ 一定, } y \sim x & t \text{ 时刻各点分布} \end{cases}$

物体弹性变形

线变 切变 体变 $\rightarrow$ 模量

$$\frac{F}{S} = E \frac{\Delta l}{l}$$

量纲一致

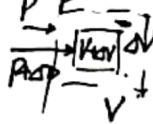


$$\frac{F}{S} = G \frac{\Delta \theta}{\theta} = G \gamma$$



$$\Delta p = K \frac{\Delta V}{V}$$

$$K = \frac{1}{k} = \frac{1}{V} \frac{\Delta V}{\Delta p}$$



仅固体能承载纵波

Δx 段 平均应变

$$\frac{y(x+\Delta x, t) - y(x, t)}{\Delta x}$$

应变 $\frac{\partial y}{\partial x}$ 应力 $\frac{F(x, t)}{S}$

应力 = 模量 $\times$ 应变

$$f(t - \frac{x}{u}) \quad (+x \text{ 向})$$

$$f(t + \frac{x}{u}) \quad (-x \text{ 向})$$

$$y(x, t) = A \cos \omega(t - \frac{x}{u})$$

$$\text{or } y(x, t) = A \cos(\omega t - \frac{2\pi}{\lambda} x)$$

$$\phi(x) = -\frac{x}{\lambda} \cdot 2\pi$$

$$\frac{F(x, t)}{S} = E \cdot \frac{\partial y}{\partial x}$$



由牛顿定律 $\frac{F_2}{\Delta x} - \frac{F_1}{\Delta x} = \rho \cdot \frac{\partial^2 \xi}{\partial t^2}$

$$F_2 - F_1 = (\rho S \Delta x) \cdot \frac{\partial^2 \xi}{\partial t^2}$$

$$\frac{\partial^2 \xi}{\partial t^2} \rho \Delta x = \frac{F_2}{S} - \frac{F_1}{S}$$

$$\frac{\partial^2 \xi}{\partial t^2} \rho \Delta x = E \left[\left(\frac{\partial \xi}{\partial x} \right)_2 - \left(\frac{\partial \xi}{\partial x} \right)_1 \right]$$

$$\frac{\partial^2 \xi}{\partial t^2} \rho = E \frac{\partial^2 \xi}{\partial x^2}$$

$$u = \sqrt{\frac{E}{\rho}} \quad (\text{波速})$$

任一以 $t \mp \frac{x}{u}$ 为变量的波函数

$y = f(t \mp \frac{x}{u})$ 都是波动方程解

弦的波动方程. 波速



$$T_1 \cos \theta_1 = T_2 \cos \theta_2 = F$$

$$\frac{T_2 \sin \theta_2}{T_2 \cos \theta_2} - \frac{T_1 \sin \theta_1}{T_1 \cos \theta_1} = \Delta m \cdot \left(\frac{\partial^2 \xi}{\partial t^2} \right)_x$$

$$\frac{F}{\cos \theta_2} \sin \theta_2 - \frac{F}{\cos \theta_1} \sin \theta_1 = \Delta m \cdot \left(\frac{\partial^2 \xi}{\partial t^2} \right)_x$$

$$F \left[\left(\frac{\partial y}{\partial x} \right)_{x+\Delta x} - \left(\frac{\partial y}{\partial x} \right)_x \right] = \Delta m \cdot \left(\frac{\partial^2 y}{\partial t^2} \right)_x$$

$$F \cdot \frac{\partial^2 y}{\partial x^2} \cdot \Delta x = \Delta m \left(\frac{\partial^2 y}{\partial t^2} \right)_x$$

$$u = \sqrt{\frac{F}{\rho}} \quad (\text{弹性绳上横波})$$

气体中 $u = \sqrt{\frac{\gamma R T}{M}}$ γ - 绝热比

液体 $u = \sqrt{\frac{k}{\rho}}$ k - 体积模量

固体 $\begin{cases} u_{\text{横}} = \sqrt{\frac{G}{\rho}} & G = \frac{F/S}{\phi} \quad (\text{切变模量}) \\ u_{\text{纵}} = \sqrt{\frac{E}{\rho}} & E = \frac{F/S}{\Delta l/l} \quad (\text{杨氏模量}) \end{cases}$

$$u_{\text{纵}} > u_{\text{横}}$$

波的能量

形变势能

$$\Delta W_p = \int_0^{y_{\text{max}}} F dy$$

$$= \int_0^{y_{\text{max}}} G \phi S dy$$

$$p \approx \text{平均} = \frac{1}{2} G \phi \int_0^{y_{\text{max}}} G \phi \Delta x dy$$

$$= \frac{1}{2} G \phi^2 \Delta V$$

$$E \frac{\partial^2 \xi}{\partial x^2} = \rho \cdot \frac{\partial^2 \xi}{\partial t^2}$$

$$\frac{\partial^2 \xi}{\partial x^2} = \frac{1}{u^2} \frac{\partial^2 \xi}{\partial t^2} \quad (\text{一维波动方程})$$

$$u = \sqrt{\frac{F}{\rho}} \quad (F \text{ 是元波丁绳的张力})$$

$$\Delta W_p = \frac{1}{2} G \cdot \left(\frac{\Delta y}{\Delta x} \right)^2 \Delta V$$

$$= \frac{1}{2} u^2 \rho \cdot \left(\frac{\Delta y}{\Delta x} \right)^2 \Delta V$$



扫描全能王 创建

全反射的临界角

$$\sin i_c = \frac{n_2}{n_1}$$

全反射时界面附近有透射波

光纤纤维 - 中央折射率大, 表层折射率小, 波的叠加, 驻波

波的独立传播原理

波的相遇处, 质元的位移等于各波单独在该处产生位移的叠加.

波的干涉条件 $\begin{cases} \text{同} \\ \text{振动方向同} \\ \Delta\phi \text{ 固定} \end{cases}$

驻波

$$\begin{cases} +x & y_1 = A_1 \cos(\omega t - \frac{x}{\lambda} 2\pi) \\ -x & y_2 = A_2 \cos(\omega t + \frac{x}{\lambda} 2\pi) \end{cases}$$

$$y = y_1 + y_2$$

特点: ① 振幅, 各处不等大, 有波腹 ($A_{\max}$)
波节 ($A_{\min}$)

② 相位, 不传播, 分段振动

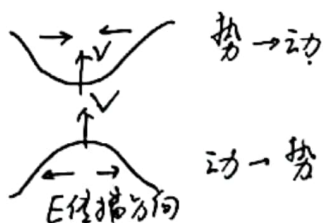
$$2A \cos \frac{x}{\lambda} 2\pi \quad x = \frac{\lambda}{4}, \frac{3}{4}\lambda, \dots$$

在波节两侧变节向

③ 能量, 合能流密度 ($A_1 = A_2$)

$$\vec{\omega} \cdot \vec{u} + (-\vec{u}) \cdot \vec{\omega} = 0$$

平均无能量传播但各质元间有能量交换



机械波入射, 界面关系

① 界面两侧质元位移同 (接触)

$$[y + y']_{x=0} = [y_2]_{x=0}$$

② 界面两侧张力相同 (牛顿定律)

$$[\frac{F_1}{S} + \frac{F_1'}{S}]_{x=0} = [\frac{F_2}{S}]_{x=0}$$

$$E_1 [\frac{\partial y_1}{\partial x} + \frac{\partial y_1'}{\partial x}]_{x=0} = E_2 [\frac{\partial y_2}{\partial x}]_{x=0} \text{ (纵波)}$$

$$(A_1 = A_2)$$

$$y = \frac{2A \cos \frac{x}{\lambda} 2\pi \cdot \cos \omega t}{|2A \cos \frac{x}{\lambda} 2\pi| = A', \text{ 无 } x}$$

$$|2A \cos \frac{x}{\lambda} 2\pi| = A', \text{ 无 } x$$

$A_1 \neq A_2$ 时, 若 $A_2 = A_1 + \Delta A > A_1$

$$y = \frac{2A_1 \cos \frac{x}{\lambda} 2\pi \cos \omega t}{\text{典型驻波}} + \frac{\Delta A \cos \omega t + \frac{x}{\lambda} 2\pi}{\text{行波}}$$

典型驻波

行波

可称驻波, 但波节有振动

相位关系

反射波 $\begin{cases} \text{① } Z_1 > Z_2, \phi_1' = \phi, \text{ 反射波与入射波同相} \\ \text{② } Z_1 < Z_2, \phi_1' = \phi + \pi, \text{ 反相} \end{cases}$

透射波 无论 Z_1, Z_2 关系 同相



扫描全能王 创建

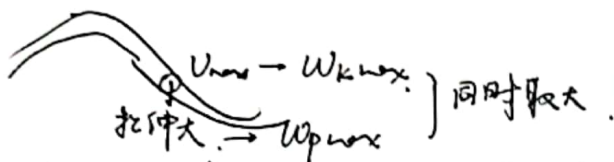
$$\frac{\Delta y}{\Delta x} \approx \frac{\partial y}{\partial x}$$

$$y = A \cos(\omega t - \frac{x}{u})$$

$$y' = \omega A \sin(\omega t - \frac{x}{u}) \cdot \frac{1}{u}$$

振动系统 $E_k \neq E_p$ $E_k + E_p = \text{const}$

波动质元 $E_k \rightarrow E_p$ $\Delta W_k + \Delta W_p \neq \text{const}$
 $\Delta W_k = \Delta W_p$ (与同介质交换能量)



能流密度 S 单位时间垂直于波线向单位面积的能量

波强度 $I = \overline{S}$

Z - 特性阻抗 Z 大 波密介质

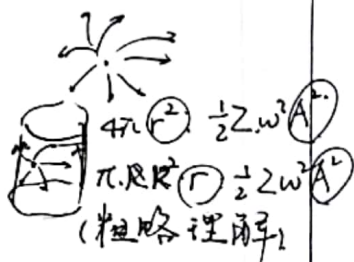
利用 $I = \frac{1}{2} Z \omega^2 A^2$ 能量守恒

对平面波 $A = \text{const}$

球面波 $A r = \text{const}$

柱面波 $A \sqrt{r} = \text{const}$

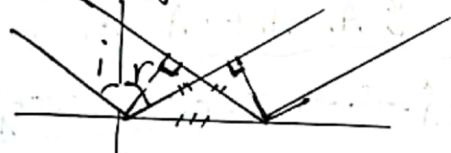
r - 到波源距离



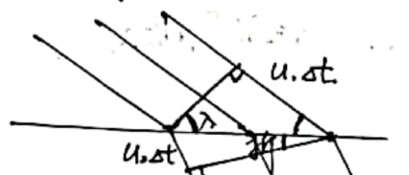
惠更斯原理 (略)

衍射. 波长长. 衍射明显

反射. 折射



三角形全等 → 反射定律 $i=r$



$$\frac{\sin i}{\sin r} = \frac{u_1}{u_2} = \frac{n_2}{n_1} = \text{const.}$$

$$(u = \frac{c}{n}) \quad n_{21} = \frac{n_2}{n_1} \text{ 相对折射率}$$

$$\Delta W_p = \frac{1}{2} \rho \Delta V \omega^2 A^2 \sin^2 \omega(t - \frac{x}{u})$$

$$\Delta W_k = \frac{1}{2} \rho \Delta V (\frac{\partial y}{\partial t})^2$$

$$\therefore \Delta W = \rho \Delta V \omega^2 A^2 \sin^2 \omega(t - \frac{x}{u})$$

$$\overline{W} = \rho \omega^2 A^2 \sin^2 \omega(t - \frac{x}{u}) \propto \omega^2 A^2$$

特征量

$$\overline{W} = \frac{1}{2} \rho \omega^2 A^2 \text{ 适用于弹性波}$$

$$I = \overline{W} u = \frac{1}{2} \rho \omega^2 A^2 u$$

$$\underline{Z = \rho u} \quad \frac{1}{2} Z \omega^2 A^2$$

$$\frac{\sin i}{\sin r} = \frac{n_2}{n_1}$$



振幅关系 $I = \frac{1}{2} \rho \omega^2 A^2 u = \frac{1}{2} Z \omega^2 A^2$

$$A_1' = \left| \frac{Z_1 - Z_2}{Z_1 + Z_2} \right| A_1$$

$$A_2 = \frac{2Z_1}{Z_1 + Z_2} A_1$$

参照弹性碰撞
公式

$\Delta Z_1, Z_2$ 互换, R, T 不变

$\Delta Z_1 \gg Z_2, Z_2 \gg Z_1, R \approx 1$ 全反射

$\Delta Z_1 \approx Z_2$ 全透射

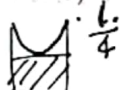
简正模式

两端固定的弦 须满足 $n \cdot \frac{\lambda_n}{2} = L$
($n \in \mathbb{Z}$)

$$v_n = \frac{u}{\lambda_n} = n \cdot \frac{u}{2L} \quad \text{固有频率}$$

$$\text{重申 } u = \sqrt{\frac{F}{\rho}} \quad (\text{弹性弦})$$

$\Rightarrow$ 克拉尼图形

有水的升  $\frac{L}{4}$

弦 手不按时 发基音

解: ① $L = \dots n \lambda_n = \dots n \cdot \frac{u}{v}$

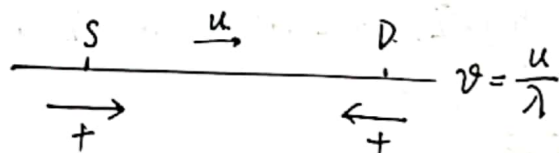
② $A \cos \omega(t \pm \frac{x}{u})$ 有时可 ω

$$A \cos(\omega t \pm \frac{2\pi x}{\lambda})$$

或 $A \cos 2\pi(\frac{t}{T} \pm \frac{x}{\lambda})$

多普勒效应

波源 S 观测者 D



① $v_D \neq 0$

$$v' = \frac{u + v_D}{\lambda} = \frac{u + v_D}{u} v$$

$$v_D \leftarrow$$

② $v_S \neq 0$

$$v' = \frac{u}{\lambda'} = \frac{u}{(u - v_S)T} = \frac{u}{u - v_S} v$$

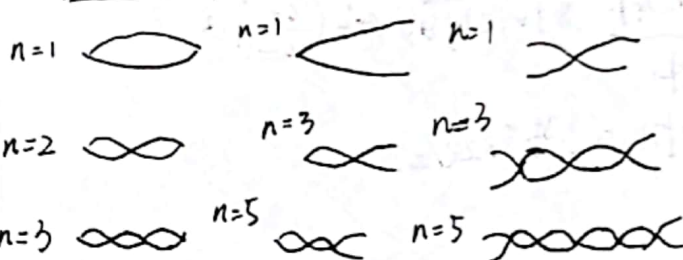
⑤

$$\text{反射比 } R = \frac{I_1'}{I_1} = \left(\frac{A_1'}{A_1} \right)^2 = \left(\frac{Z_1 - Z_2}{Z_1 + Z_2} \right)^2$$

$$\text{透射比 } T = \frac{I_2}{I_1} = \frac{Z_2}{Z_1} \left(\frac{A_2}{A_1} \right)^2 = \left(\frac{2Z_1}{Z_1 + Z_2} \right) \cdot \frac{Z_2}{Z_1} = \frac{4Z_1 Z_2}{(Z_1 + Z_2)^2}$$

$$R + T = 1 \quad (\text{能量守恒, 参照弹性})$$

$L = \dots \frac{\lambda}{2}$ ① 两端定 $\frac{\lambda}{2}$ ② 单端定 $\frac{\lambda}{4}$ ③ 中间定 $\frac{\lambda}{2}$



固定处必是波节

③ $v_D \neq 0, v_S \neq 0$

$$v' = \frac{u + v_D}{u - v_S} v$$

加观 (分子)

减源 (分母)

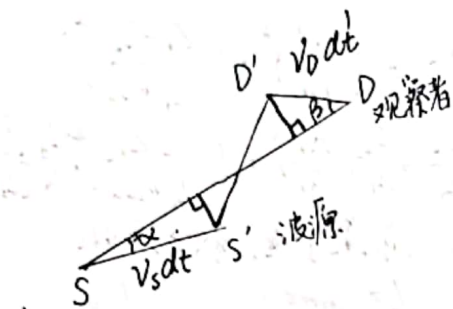
S 波源运动 作用在 λ 上 $\lambda' = \frac{(u - v_S)}{u} \lambda$

D 观测者运动 作用在 u 上 $u' = \frac{u + v_D}{u}$



扫描全能王 创建

多普勒效应. 普遍公式.



静S

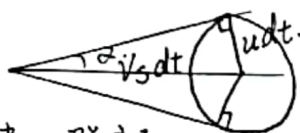
D反射

$$S \text{ 接收到的 } v' = \frac{u+v}{u-v} u$$

相对

相对论与多普勒效应.

马赫锥 (当 $v_s > u$ 时)



复波. 群速度

不同波叠加. 非同频波.

波群. 波包. 传播速度 $u_g \rightarrow$ 群速度.

孤子. 形状不变的孤立波 (波包的两种作用抵消).

声波.

$$P = P_0 + \Delta P = \frac{\partial y}{\partial x} dx$$

$$\Delta P = \frac{\partial P}{\partial x} dx$$

牛顿定律

$$S(P + \frac{\partial P}{\partial x} dx) - SP = \rho \cdot S dx \cdot \frac{\partial^2 y}{\partial t^2}$$

$$\frac{\partial P}{\partial x} dx = \rho dx \cdot \frac{\partial^2 y}{\partial t^2}$$

$$\Delta V = \frac{S \cdot \frac{\partial y}{\partial x} dx}{S dx} = \frac{\partial y}{\partial x}$$

体变公式

$$\Delta P = k \cdot \frac{\Delta V}{V} = k \cdot \frac{\partial y}{\partial x}$$

$$\frac{\partial(\Delta P)}{\partial x} = k \cdot \frac{\partial^2 y}{\partial x^2} = \rho \cdot \frac{\partial^2 y}{\partial t^2} (*)$$

$$u = \sqrt{\frac{k}{\rho}}$$

$$k \cdot \frac{\partial^2 P}{\partial x^2} = \rho \frac{\partial^2 P}{\partial t^2}$$

$$u' = \sqrt{\frac{k}{\rho}}$$

$$\Delta P = -k \frac{\partial y}{\partial x} = -k \cdot \frac{u}{A} A \sin \omega t - \frac{x}{\lambda} = -\rho u \omega A \sin \omega t - \frac{x}{\lambda}$$

DS连线与 V_0 夹角

$$\text{普遍公式 } \frac{v'}{v} = \frac{dt}{dt'} = \frac{u + v_o \cos \beta}{u - v_s \cos \alpha}$$

DS连线与 V_s 夹角

$$\frac{v'}{v} = \sqrt{\frac{c+v}{c-v}} = \sqrt{\frac{1+\beta}{1-\beta}} \quad (\text{纵向})$$

$$\frac{v'}{v} = \sqrt{1-\beta^2} \quad (\text{横向})$$

$$\sin \alpha = \frac{u}{v_s} \quad \frac{v_s}{u} : \text{马赫数}$$

$$u_g = \frac{d\omega}{dk} \quad u = \frac{\omega}{k} \quad u_g = u - \lambda \frac{du}{d\lambda}$$

无色散介质 $\frac{du}{d\lambda} = 0 \quad u_g = u$

色散 $\rightarrow$ 波包打散 群速度无意义

非线性介质 $\rightarrow$ 波包拥挤

y 代表那些满足波动方程. 即以波的形式传播的物理量. 因而如果一个物理量满足(*)的形式. 则

声压振幅

$$P_m = \rho u A \omega$$

声强

$$I = \frac{1}{2} \rho \omega^2 A^2 u = \frac{P_m^2}{2 \rho u}$$

声级

$$L = 10 \lg \frac{I}{I_0} \text{ Bel} = 10 \lg \frac{I}{I_0} \text{ dB}$$



扫描全能王 创建

大学物理B11. 狭义相对论

一切物理规律在所有惯性系中形式相同。

光速不变

△同时的相对性 ← 直接结果

事件：与参考系无关

Einstein 火车

1. 效应是相对的。

位于事件同时发生的那个参考系下后方的事件先发生。

2. 路径不同导致效应。

洛伦兹变换。

时间膨胀 长度收缩 (原时最短, 原长最长)

x 系:  d
 $d' = \frac{1}{2} u \Delta t'$
 $\Delta t' = \frac{2d}{u}$
 x 系:  l
 $l^2 = d^2 + (u \cdot \frac{\Delta t'}{2})^2$
 $\Delta t = \frac{2l}{c}$

x 系 $\Delta t = 0$ 同时测两端



时空间隔不变性。

时间间隔 $\Delta t = t_2 - t_1$

$\Delta t' = t'_2 - t'_1$

空间间隔 $\Delta l = \sqrt{(\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2}$

$\Delta l' = \dots$

四维时空点 $\vec{s} = (x, y, z, ict)$

闵可夫斯基空间

对有因果关系事件 $|\Delta s|^2 < 0$ 类时区。



$(\Delta l < ct)$

解题. 注意同时考虑长度的变化与同时的相对性。

①

正变换

$$\begin{cases} x' = \frac{x - ut}{\sqrt{1 - \frac{u^2}{c^2}}} \\ y' = y \\ z' = z \\ t' = \frac{t - \frac{u}{c^2}x}{\sqrt{1 - \frac{u^2}{c^2}}} \end{cases}$$

u. 参考系间速度。
带'的'为在
被观测一方参考
系内测得的结果

逆变换

$$\begin{cases} x = \frac{x' + ut'}{\sqrt{1 - \frac{u^2}{c^2}}} \\ y = y' \\ z = z' \\ t = \frac{t' + \frac{u}{c^2}x'}{\sqrt{1 - \frac{u^2}{c^2}}} \end{cases}$$

$\beta = \frac{u}{c}$
 $\gamma = \frac{1}{\sqrt{1 - \frac{u^2}{c^2}}}$

$\Delta t = \frac{\Delta t'}{\sqrt{1 - \frac{u^2}{c^2}}} = \gamma \Delta t'$ 原时
 $(\Delta x' = 0)$

$\Delta x' = \frac{\Delta x - u \Delta t}{\sqrt{1 - \frac{u^2}{c^2}}} = \frac{\Delta x}{\sqrt{1 - \frac{u^2}{c^2}}}$

$\Rightarrow l = \sqrt{1 - \frac{u^2}{c^2}} l'$ 原长

$(\Delta s)^2 = (\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2 + \cancel{c^2 \Delta t^2} - (c \Delta t)^2$
 $= (\Delta s')^2 = \dots$ ($i^2 = -1$)



扫描全能王 创建

速度变换 在S'系中速度
 $v_x = \frac{dx}{dt}$ $(v_x)' = \frac{dx'}{dt'}$ 参考系
 $x' = \gamma(x - ut)$ $t' = \gamma(t - \frac{u}{c^2}x)$
 解题: S'系中若V'与u不平行
 采用分别算 V_x V_y 方式.

正变换

$$\begin{cases} v_x' = \frac{dx'}{dt'} = \frac{v_x - u}{1 - \frac{u}{c^2}v_x} \\ v_y' = \frac{v_y}{1 - \frac{u}{c^2}v_x} \sqrt{1 - \frac{u^2}{c^2}} \\ v_z' = \frac{v_z}{1 - \frac{u}{c^2}v_x} \sqrt{1 - \frac{u^2}{c^2}} \end{cases}$$

逆变换

$$\begin{cases} v_x = \frac{v_x' + u}{1 + \frac{u}{c^2}v_x'} \\ v_y = \frac{v_y'}{1 + \frac{u}{c^2}v_x'} \sqrt{1 - \frac{u^2}{c^2}} \\ v_z = \frac{v_z'}{1 + \frac{u}{c^2}v_x'} \sqrt{1 - \frac{u^2}{c^2}} \end{cases}$$

多普勒 $f = f_0 \sqrt{1 - \frac{v^2}{c^2}}$

一致性

$$\omega = \frac{\omega_0}{\gamma(1 - \frac{v}{c} \cos \theta)}$$

相对论质量

$$m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}} = \gamma m_0$$

$$\begin{cases} F_n = m a_n = \gamma m_0 a_n = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}} a_n \\ F_t = m a_t + v \frac{dm}{dt} = \gamma^3 m_0 a_t = \frac{m_0}{(\sqrt{1 - \frac{v^2}{c^2}})^{3/2}} a_t \end{cases}$$

(1) $\vec{a} \times \vec{F}$
 (2) 切向加速比法向加速困难.

$$\vec{a} = \frac{\vec{F}}{m(v)} - \frac{\vec{v}(\vec{v} \cdot \vec{F})}{m(v)^3 c^2}$$

(1) $v \ll c$ $\vec{a} = \frac{\vec{F}}{m}$
 (2) $\vec{v} \perp \vec{F}$ $\vec{F} = \gamma m_0 \vec{a}$

动能

$$E_k = m_0 c^2 \left(\frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} - 1 \right)$$

$$E = m c^2 \quad E_0 = m_0 c^2 \text{ 静能量}$$

$$E_k = m c^2 - m_0 c^2$$

$$\Delta m = \sum \Delta m_{\text{初}} - \sum \Delta m_{\text{末}}$$

释能效率 $\frac{\Delta E}{E_0}$

相对论多普勒效应.

相对论质量.

动力学方程要求

① $v \ll c$ 时返回牛顿方程

力 = 动量对时间变化率

② 满足 Einstein 相对论原理.

守恒关系 $\vec{F} = \frac{d\vec{p}}{dt}$

及定义 $\vec{p} = m\vec{v}$, $m = m(v)$

由 $\begin{cases} v_z = \frac{v_z' + u}{1 + \frac{u}{c^2}v_z'} \\ M = m_1 + m_2 \\ Mv = m_1 v_1 + m_2 v_2 \end{cases}$

令 $v_z = 0 \Rightarrow$

力与加速度 $\Rightarrow$

$\vec{F} = \frac{d\vec{p}}{dt}$, $\vec{p} = m\vec{v}$

$\frac{d\vec{v}}{dt} = a_n \hat{n} + a_t \hat{t}$, $m = m(v)$
 $\vec{F} = \frac{dm}{dt} \vec{v} + m(a_n \hat{n} + a_t \hat{t})$
 法. 切.

解题: $F = m \frac{v}{R}$ 等规律一致 (动能定理),
 只是 $m \rightarrow m(v)$

动能定理.

$$dE_k = mv dv + v^2 dm = c^2 dm$$

相对论能量.

当过程 (如核反应) 前后可将系统看成质点集时

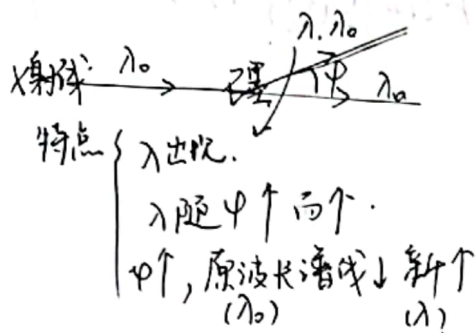


扫描全能王 创建

$$\begin{cases} E = mc^2 \\ m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}} \Rightarrow \\ p = m\vec{v} \end{cases}$$

解题: E 包括一切能量.

康普顿散射.



$$\Delta\lambda = \lambda - \lambda_0 \text{ 只与 } \varphi \text{ 有关} \\ (\lambda > \lambda_0)$$

光子与静止电子的弹性碰撞.

光子 $E \downarrow \lambda \uparrow$

动量-能量变换.

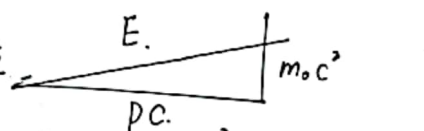
法一:

$$\vec{p}' = m\vec{v}' \quad E' = mc^2 \cdot (E' = E)$$

V 变换公式

法二: 对 mc^2 ,

$m_0 c^2$ 不变量



$$E^2 - p^2 c^2 = m_0^2 c^4 = E'^2 - p'^2 c^2$$

$$\text{类似 } (\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2 - (c\Delta t)^2 = \Delta s^2 = \text{const}$$

$$\frac{E^2}{c^2} \sim t^2$$

$$p^2 = p_x^2 + p_y^2 + p_z^2$$

$$p_x \sim x \quad (t, x, y, z) \rightarrow (t', x', y', z')$$

$$p_y \sim y \quad (\frac{E}{c}, p_x, p_y, p_z) \rightarrow (\frac{E'}{c}, p'_x, p'_y, p'_z)$$

$$p_z \sim z$$

多普勒效应.

□ R (S 系中静止)

S' 系中静止 $\vec{v} = \vec{v}'$ (能量守恒).

$$S \xrightarrow{u} S'$$

$$E^2 = p^2 c^2 + m_0^2 c^4$$

$$\text{代入 } E = E_k + m_0 c^2$$

$$\text{有 } E_k^2 + 2E_k \cdot m_0 c^2 = p^2 c^2$$

光子 $m_0 = 0$

$$E = pc$$

实验规律.

$$\Delta\lambda = \lambda_c (1 - \cos\varphi) = 2\lambda_c \sin^2 \frac{\varphi}{2}$$

λ_c : 电子的康普顿波长

$$\tan\varphi = \frac{p \sin\theta}{p_0 - p \cos\theta} = \frac{1}{(1 + \lambda_c/\lambda_0) \tan \frac{\theta}{2}}$$

光子动量 (散射后)
 电子动量
 光子动量
 失角

$$t' = \frac{t - \frac{u}{c^2} x}{1 - \frac{u^2}{c^2}} \rightarrow \frac{E'}{c^2} = \frac{\frac{E}{c^2} - \frac{u}{c^2} p_x}{1 - \frac{u^2}{c^2}}$$

$$\begin{cases} E' = \gamma \cdot (E - \frac{u}{c^2} p_x) \\ x' = \frac{x - ut}{1 - \frac{u^2}{c^2}} \end{cases} \rightarrow \begin{cases} E' = \gamma \cdot (E - u p_x) \\ p'_x = \frac{p_x - u \cdot \frac{E}{c^2}}{1 - \frac{u^2}{c^2}} \end{cases}$$

$$\begin{cases} y' = y \\ z' = z \end{cases} \rightarrow \begin{cases} p'_y = p_y \\ p'_z = p_z \end{cases}$$

$$\vec{v} = \frac{\sqrt{1 - \frac{u^2}{c^2}}}{1 - \frac{u}{c} \cos\theta} \cdot \vec{v}_0$$

横向 纵向 作用 θ .



扫描全能王 创建

热力学第一定律

内能 $E_2 - E_1 = A_{\text{绝热}}$

绝热过程 $Q=0$

$\Delta T_{\text{绝热}} > 0$ 准静态

L 容器长度

$T = \frac{L}{v}$ 分子热运动速率

摩尔热容

$C_v, C_p, C_p > C_v$

$C_{v,m} < C_{p,m}$

等压过程

$dQ_p = dE + dA_p$ (热-)

$dQ_p = \nu C_{p,m} dT$

$dE = \nu C_{v,m} dT$

$dA = p dV = \nu R dT$

$dE = \nu \cdot \frac{1}{2} R dT$

$\nu dT = \nu C_{v,m} dT$

绝热过程 $dQ=0$

$dE = -dA$

理想气体状态方程
 $R = C_{p,m} - C_{v,m}$
热一律

$Q = (E_2 - E_1) + A$

$= \Delta E + A$

$A > 0$ 对外做功

$Q > 0$ 吸热

$\Rightarrow dQ = dE + dA$

$C_v = \left(\frac{dQ}{dT} \right)_v$

$C_p = \left(\frac{dQ}{dT} \right)_p$

$C_{v,m} = \frac{1}{\nu} \left(\frac{dQ}{dT} \right)_v, C_{p,m} = \frac{1}{\nu} \left(\frac{dQ}{dT} \right)_p$

理想气体内能公式

$\Delta E = \nu \cdot C_{v,m} \Delta T$

$C_{p,m} - C_{v,m} = R$

$r = \frac{C_{p,m}}{C_{v,m}}$

$\begin{cases} C_{v,m} = \frac{1}{2} R \\ C_{p,m} = (\frac{1}{2} + 1) R \end{cases} \Rightarrow r = \begin{cases} \frac{5}{3} & \text{单分子} \\ \frac{7}{5} & \text{双分子} \\ \frac{8}{6} = \frac{4}{3} & \text{三分子} \end{cases}$

理想气体 · 绝热过程 (准静态)

$P_1 V_1^r = P_2 V_2^r$

$T_1 V_1^{r-1} = T_2 V_2^{r-1}$

$P_1^{1-r} T_1^{-r} = P_2^{1-r} T_2^{-r}$

理想气体过程	特征	参量关系	Q	A	ΔE
等体	$V = \text{const.}$	$P/T = C$	$\nu C_{v,m} \Delta T$	0	$\nu C_{v,m} \Delta T$
等压	$P = \text{const.}$	$V/T = C$	$\nu C_{p,m} \Delta T$	$P \Delta V = \nu R \Delta T$	$\nu C_{v,m} \Delta T$
等温	$T = \text{const.}$	$PV = C$	$\nu RT \ln V_2/V_1$ $\nu RT \ln P_1/P_2$	$\nu RT \ln V_2/V_1$ $\nu RT \ln P_1/P_2$	0
绝热	$dQ=0$	$PV^r = C$ $T V^{r-1} = C$ $P^{1-r} T^{-r} = C$	0	$-\nu C_{v,m} \Delta T$ $\frac{P_2 V_2 - P_1 V_1}{1-r}$	$\nu C_{v,m} \Delta T$



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